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IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Date of publication xxxx 00, 0000, date of current version xxxx 00, 0000.

Digital Object Identifier 10.1109/ACCESS.2024.0429000

Aspect-based Sentiment Analysis in Ho Chi Minh city Hotel Reviews using Aspect-Conditioned BiLSTM with Cross-Aspect Attention and Ordinal Regression

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ABSTRACT The rapid growth of tourism in Ho Chi Minh City has generated a huge volume of Vietnamese hotel reviews, creating an urgent demand for effective aspect-based sentiment analysis (ABSA). This paper introduces a new large-scale benchmark dataset of 20,290 manually annotated reviews collected from Agoda, Google Reviews, and Traveloka, covering ten hotel aspects with 5-point ordinal sentiment labels (-2 to +2). The dataset is publicly available at https://github.com/tamkhojj/AC-BiLSTM represents the large and comprehensive Ho Chi Minh city tourism ABSA resource to date. We further propose AC-BiLSTM, a novel model that combines PhoBERT embeddings with aspect-conditioned BiLSTM, multi-head aspect attention, cross-aspect attention, aspect-conditioned gating, and ordinal regression loss. Extensive experiments show that AC-BiLSTM significantly outperforms strong LSTM-based baselines, achieving a macro F1-score of 0.8899, Precision of 0.8921, Recall of 0.8880, and Accuracy of 0.9456. Ablation studies and error analysis confirm the effectiveness and robustness of the proposed components. The dataset are publicly released to facilitate future research on low-resource Vietnamese ABSA.

INDEX TERMS Aspect-based sentiment analysis, BiLSTM, Aspect-Condition Gating, Low Resource Dataset, Sentiment Classification.

I. INTRODUCTION

Vietnam has emerged as a vibrant hub in the global tourism landscape, driven by its rich cultural heritage, diverse natural attractions, and strategic economic policies aimed at fostering sustainable growth. According to recent statistics from the Vietnam National Administration of Tourism [1], the country welcomed over 12.6 million international visitors in the first nine months of 2025, reflecting a robust recovery and a 45% increase from the previous year. This resurgence is particularly pronounced in urban centers and coastal regions, where tourism infrastructure has been significantly enhanced to accommodate growing demand. Among these destinations, Ho Chi Minh City (HCMC), Vietnam's economic and cultural epicenter, stands out as a leading contributor to this upward trajectory. As reported by the municipal Department of Tourism [4], HCMC attracted over 6.58 million international visitors and 33.08 million domestic tourists in the first ten months of 2025, generating a total revenue of VND 208.06 trillion (approximately USD 8.32 billion), which accounts for 71.7% of the year's target of VND 290 trillion. This growth, bolstered by strategic initiatives such as visa exemptions for 12 European countries effective since August 15, 2025, and the promotion of medical and wellness tourism in collaboration with the Mekong Delta and Southeastern regions, underscores HCMC's pivotal role in shaping Vietnam's tourism industry [2], [3]. The city's public transport system further supports this influx, serving 55.1 million passengers in October alone, with the Bến Thành–Suối Tiên metro line carrying 1.8 million and air transport handling 3.3 million passengers. These developments not only highlight HCMC's readiness for year-end peak seasons and international events but also emphasize the need for data-driven strategies to sustain momentum and enhance visitor experiences.

The hospitality sector in HCMC, comprising thousands of hotels and accommodation facilities, plays a critical role in sustaining this momentum. With the city preparing for a peak season and hosting large international delegations, online reviews have become an indispensable resource for both travelers and service providers. These reviews, sourced from platforms such as Agoda, Google Reviews, and Traveloka, of-

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fer detailed insights into guest experiences, influencing booking decisions and guiding service improvement strategies. Indeed, prior work has demonstrated that incorporating aspect categorization into hotel recommender systems can significantly enhance recommendation accuracy and user satisfaction [33]. However, the sheer volume and linguistic diversity of these reviews pose significant challenges for traditional sentiment analysis approaches, particularly in a low-resource language context like Vietnamese. Conventional sentiment analysis techniques, which typically operate at document or sentence levels, often fail to capture the nuanced opinions embedded within specific aspects of a review, such as room quality, service efficiency, or pricing fairness. For instance, a single review might praise the cleanliness of a hotel while critiquing its location, necessitating a more granular analytical framework to discern these multifaceted sentiments accurately. Moreover, Vietnamese reviews frequently incorporate slang, idiomatic expressions, and cultural nuances that are difficult for non-native models to interpret, exacerbating the need for specialized tools tailored to the local context.

Aspect-Based Sentiment Analysis (ABSA) addresses this limitation by associating sentiments with specific attributes or aspects mentioned in the text, thereby enabling a fine-grained understanding of customer feedback. Introduced as a competitive task in SemEval-2014 and further refined through subsequent challenges like SemEval-2016, ABSA encompasses subtasks such as Aspect Term Extraction (ATE), Aspect Term Polarity (ATP), Aspect Category Detection (ACT), and Aspect Category Polarity (ACP) [5]–[7]. This approach has gained traction in the tourism domain, leveraging multilingual and multi-domain datasets to enhance service quality and maintain competitiveness. Recent advancements in deep learning, particularly with Bidirectional Long Short-Term Memory (BiLSTM) networks and Transformer-based models like BERT, have further improved ABSA performance across various languages. For example, in Indonesian hotel reviews, deep learning models have achieved accuracies exceeding 90% [8], while BiLSTM-based approaches on tourist attraction reviews reported strong results with F1-scores around 0.85 [9]. In Arabic, LSTM networks applied to reviews yielded robust sentiment classification with accuracies up to 85% [10]. More recently, hybrid stacking ensembles on Kazakhstan tourist reviews demonstrated superior performance in handling multilingual data [11], and the UzABSA dataset for Uzbek language introduced fine-grained annotations for under-resourced settings [12]. Traditional machine learning models have also been effective in domain-specific applications, such as aspect analysis of Bangkok's public transportation reviews, achieving up to 82% accuracy [13]. These efforts highlight the global applicability of ABSA in tourism, yet they predominantly focus on higher-resource languages or English-dominant datasets, leaving low-resource languages like Vietnamese underrepresented. Surveys of ABSA datasets, including the recent systematic review by Hua et al. [32], reveal that while over 98 publicly available corpora exist across 25 domains, only a fraction address non-English tourism contexts, with limited coverage for Southeast Asian languages [14], [15]. This scarcity underscores the challenges in developing reliable ABSA systems for linguistically diverse regions like Vietnam, where cultural idioms and code-mixing further complicate analysis.

This study seeks to bridge this research gap by introducing novel contributions to the field of ABSA tailored to HCMC hotel reviews in Vietnamese, a low-resource language. Our work emphasizes the development of resources and models that can handle the unique linguistic characteristics of Vietnamese tourism feedback, providing actionable insights for stakeholders in a rapidly growing sector. Specifically, our contributions are as follows:

- A comprehensive new dataset comprising 20,290 user-generated hotel reviews from HCMC, meticulously annotated across ten distinct aspects—room, service, location, price, food and beverage, amenities, cleanliness, transportation, policy, and others—with ordinal sentiment scores ranging from -2 (strongly negative) to +2 (strongly positive). This dataset addresses the critical lack of annotated resources for Vietnamese, a low-resource language, enabling robust training and evaluation of ABSA models in tourism contexts previously underexplored due to data scarcity.
- The Aspect-Conditioned BiLSTM with Cross-Aspect Attention and Ordinal Regression (AC-BiLSTM) model, which innovatively integrates PhoBERT for contextual embeddings tailored to Vietnamese, BiLSTM for sequential processing, multi-head attention for capturing aspect-specific sentiments, and contrastive learning for refining ordinal predictions. This approach not only outperforms other methods with an F1-score of 0.8899 and Accuracy, Precision, and Recall of 0.9456, 0.8921, and 0.8880, respectively, but also demonstrates strong robustness to slang and cultural expressions, thereby advancing ABSA applicability in low-resource settings.

II. RELATED WORK

To provide a comprehensive foundation for our research, we begin by exploring the existing literature on Aspect-Based Sentiment Analysis (ABSA) within the tourism domain, followed by a focused examination of ABSA efforts specific to the Vietnamese language. Subsequently, we delve into the evolution and applications of Long Short-Term Memory (LSTM) approaches, which form a critical component of our proposed methodology. This structured review not only highlights the advancements made in these areas but also identifies key gaps that our study aims to address through the introduction of the Aspect-Conditioned BiLSTM (AC-BiLSTM) model.

A. TOURISM ASPECT-BASED SENTIMENT ANALYSIS

The application of ABSA in the tourism sector has gained significant traction due to its ability to dissect customer feedback into actionable insights across specific service attributes. One notable contribution in this area comes from Tran et al.

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[13], who applied traditional machine learning techniques to analyze 93,948 reviews of Bangkok's public transportation system. Their study focused on aspects such as cleanliness, punctuality, and staff behavior, employing models like Logistic Regression, Support Vector Machines, Decision Trees, and Random Forests.

Following this effort, the Random Forest model emerged as the most effective, achieving an impressive accuracy of 88.90%. This success underscores the continued relevance of traditional methods in handling large-scale review datasets and provides practical implications for transportation providers aiming to enhance customer satisfaction based on targeted feedback. Building on this foundation, the focus shifted to the hotel industry with Cahyaningtyas et al. [8], who explored Indonesian hotel reviews using a variety of deep learning architectures, including RNN, LSTM, GRU, BiLSTM, Attention BiLSTM, CNN, CNN-LSTM, and CNN-BiLSTM.

Their dataset covered six aspects—price, hotel, room, location, service, and restaurant—and involved a two-step classification process. The first step focused on aspect identification, while the second determined sentiment polarity (positive or negative). The LSTM model excelled in aspect classification with a 92.6% accuracy, and the CNN model achieved a 90.4% accuracy in sentiment classification when analyzed per aspect. This transition to deep learning highlights the superiority of aspect-specific analysis over holistic approaches, paving the way for more nuanced service improvements.

Further advancing this field, Af'idah et al. [9] leveraged BiLSTM on Indonesian tourist attraction reviews, addressing the sequential nature of aspect and sentiment classification. Their methodology included individual and simultaneous evaluations, with BiLSTM outperforming LSTM, recording an average accuracy of 92.22% and an F1-score of 71.06%. This improvement, driven by BiLSTM's ability to capture bidirectional context, offered a robust solution for tourism development programs in Indonesia and set the stage for exploring multilingual applications.

In the Arabic context, Al-Smadi et al. [10] introduced a sophisticated approach using BiLSTM-CRF for aspect opinion target expression extraction and a separate aspect-based LSTM for polarity classification on hotel reviews. Their dual-model strategy resulted in a 39% enhancement in aspect extraction and a 6% improvement in polarity classification compared to baseline methods. This success demonstrates the efficacy of LSTM variants in multilingual settings and encourages further exploration of hybrid techniques.

Expanding to Kazakhstan, Murzakhmetov et al. [11] proposed a hybrid stacking ensemble combining VADER, TextBlob, Stanza, and LCF-BERT on 11,454 TripAdvisor reviews. Evaluated with majority and weighted voting, this ensemble achieved an accuracy of 89.1%, precision of 83.8%, recall of 89.1%, and an F1-score of 85.2%, excelling in aspect-level sentiment detection. This approach highlights the potential of integrating diverse analytical paradigms for tourism insights in underrepresented regions and complements the multilingual focus of previous studies.

Finally, Matlatipov et al. [12] addressed the low-resource Uzbek language with the UzABSA dataset, comprising 3,500 restaurant reviews and over 6,100 sentences. Using KNN for evaluation, they reported accuracy rates between 72% and 88%, validated by Cohen's kappa and Krippendorff's α. This effort marks a significant step toward ABSA in under-resourced languages and reveals a gap in comprehensive coverage for low-resource languages like Vietnamese, which our study aims to address.

B. VIETNAMESE ASPECT-BASED SENTIMENT ANALYSIS

Turning our attention to Vietnamese, a low-resource language with unique linguistic challenges, the development of ABSA has been notably limited, prompting targeted research efforts to build foundational resources. A pioneering effort in this domain was made by Van Nguyen et al. [17], who introduced the UIT-VSFC corpus, a dataset of Vietnamese student feedback manually annotated for sentiment analysis at the sentence level.

Initial experiments with traditional machine learning models yielded reasonable performance, marking a significant step forward in Vietnamese NLP. However, the corpus's focus on academic feedback restricts its applicability to broader domains such as tourism, leading the authors to advocate for larger, more diverse datasets to improve generalizability. This limitation sets the stage for subsequent research to expand the scope of Vietnamese ABSA.

Building on this foundation, Pham et al. [18] developed the ViCloABSA dataset, consisting of 7,000 clothing reviews from Vietnamese e-commerce platforms, annotated across five aspect categories with three polarity labels. Their experiments demonstrated that large language models (LLMs) outperformed traditional methods, achieving accuracies above 85%. This progress indicates the potential of advanced models, though the dataset's narrow focus on clothing and unresolved challenges in handling slang and informal language suggest the need for more versatile applications.

To address some of these limitations, Nguyen et al. [19] advanced the field with the UIT-ViSD4SA dataset, which includes 35,396 annotated spans across 11,122 comments, designed for span detection in Vietnamese ABSA. Their BiLSTM-CRF model, enhanced with syllable, character, and XLM-RoBERTa embeddings, achieved an F1-score of 62.76%, effectively capturing complex opinion expressions. This success, however, is constrained by the dataset's size and annotation quality, indicating potential for larger-scale or semi-supervised approaches to enhance scalability.

In a business intelligence context, Phan et al. [20] developed the SA2SL system, extending ABSA to social listening using Bi-LSTM with fastText embeddings on 11,122 phone-related comments. This system recorded F1-scores of 84.48% for aspect detection and 63.06% for sentiment classification, showcasing its potential for real-time insights. Despite this achievement, it struggles with multi-label and implicit sentiments, highlighting the need for advanced models to address

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nuanced expressions in Vietnamese reviews and motivating further innovation.

Further contributing to Vietnamese NLP, Vo et al. [14] achieved an F1-score of 89.68% using PhoBERT on university feedback, demonstrating the language model's effectiveness in low-resource settings. More recently, Huynh et al. [15] analyzes Vietnamese airline customer sentiment by combining review titles and content, using a newly created dataset and multiple deep learning models. The BiLSTM-CNN model with PhoBERT embeddings achieves the best performance, with 95% accuracy and an F1-score of 0.96.

C. LSTM APPROACH FOR ASPECT-BASED SENTIMENT ANALYSIS

The evolution of Long Short-Term Memory (LSTM) networks has significantly influenced sentiment analysis, offering robust solutions for sequential data processing, which we now explore in detail to contextualize our proposed methodology. One of the early contributions in this area came from Sangeetha and Kumaran [21], who introduced the THHO-BiLSTM model, optimizing BiLSTM weights with the Taylor-Harris Hawks Optimization (THHO) algorithm.

This hybrid approach, inspired by the hunting behavior of Harris hawks and Taylor series, improved accuracy on product review datasets by systematically selecting optimal weights for hidden layers. This reduction in mean squared error and enhancement in classification robustness provided a foundation for advanced optimization techniques in sentiment analysis, inspiring further research.

Building on this, Gupta et al. [22] proposed a Hybrid Bi-GRU-LSTM model for ABSA, incorporating a three-phase process: preprocessing, enhanced aspect term extraction using TF-IDF and cosine similarity, and classification. By averaging predictions from Bi-GRU and LSTM, which respectively capture bidirectional context and long-term dependencies, the model effectively extracted aspect-specific sentiments. This balanced approach to feature extraction and sentiment classification proved effective across diverse datasets, encouraging the integration of multiple neural architectures.

Ramasamy and Elangovan [23] developed the OCas-LSTM model, integrating Seq2seq and Transformer networks with Modified Bird Swarm-Ladybug Beetle Optimization (MBS-LBO) for feature optimization. This cascaded LSTM architecture, optimized for hyperparameters, processed parallel feature extractions and fused them for ABSA. Its adaptability to complex sentiment tasks demonstrated the power of multi-network integration, paving the way for more sophisticated models.

Kokab et al. [24] combined BERT with a Convolution Bi-directional Recurrent Neural Network (CBRNN), using Dilated CNN and BiLSTM to handle noisy social media data. The Dilated CNN's varying dilation rates enabled local and global feature extraction without pooling losses, while BiLSTM captured sequential dependencies. This approach addressed out-of-vocabulary challenges with zero-shot classification, offering a robust solution for unstructured data and influencing subsequent hybrid designs.

Huang et al. [25] proposed the AEC-LSTM model, enhancing LSTM with an Emotion-enhanced LSTM (ELSTM) that included an Emotion Estimator and Modulator. Coupled with topic-level attention using Latent Dirichlet Allocation (LDA), this model modulated memory based on emotional states, improving sentiment analysis. By aligning predictions with abstract themes, it introduced a novel emotional intelligence dimension to LSTM applications, expanding the scope of contextual analysis.

Rahman et al. [26] fused RoBERTa with BiLSTM, leveraging RoBERTa's dynamic masking and enhanced pretraining for context-aware embeddings, followed by BiLSTM for long-range dependency capture. This hybrid model excelled in sentiment classification, benefiting from Transformer and RNN synergies. Its success reinforced the value of combining pre-trained models with sequential learning, a trend continued in later studies.

Tan et al. [27] developed RoBERTa-LSTM with data augmentation, using GloVe for synonym replacement and oversampling to address imbalanced datasets. The unidirectional LSTM processed augmented embeddings, focusing on lexical diversity, though lacking the bidirectional context of BiLSTM. This approach highlighted the importance of data balancing, influencing strategies to improve model performance on skewed datasets.

Prottasha et al. [28] applied Bangla-BERT with CNN-BiLSTM for low-resource Bangla, using supervised fine-tuning and a hybrid architecture to extract features and capture dependencies. This outperformed traditional embeddings like Word2Vec, demonstrating the efficacy of transfer learning in low-resource settings. Its success paved the way for similar applications in other underrepresented languages, broadening the applicability of LSTM-based models.

Kamyab et al. [29] introduced ACL-SA, combining CNN and BiLSTM with TF-IDF and GloVe embeddings, enhanced by Gaussian Noise and Dropout for robustness. The attention mechanism emphasized critical features, improving long-term dependency handling. This balanced approach to feature extraction and classification proved suitable for diverse sentiment analysis tasks, adding to the versatility of LSTM frameworks.

Khan et al. [30] utilized CNN-LSTM with ML classifiers (e.g., SVM) for Roman Urdu, using Word2Vec for embeddings and stacking LSTM layers for dependency capture. The SVM classifier on CNN-LSTM features outperformed traditional softmax, addressing low-resource language complexities. This hybrid deep learning and traditional machine learning combination underscored the potential for integrated approaches in challenging linguistic contexts.

Beyond LSTM-based architectures, other deep learning approaches have focused on explicitly modeling syntactic structures. For instance, Zhang et al. [34] introduced Aspect-specific Graph Convolutional Networks (AS-GCN) to capture long-range syntactic dependencies between aspect terms and

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context words. While effective, graph-based methods can be computationally intensive. In our work, we integrate syntactic information more directly via dependency embeddings within a BiLSTM framework to balance performance and efficiency.

These LSTM advancements highlight their versatility, yet the limited focus on cross-aspect dependencies, ordinal scales, and low-resource tourism contexts like Vietnamese ABSA reveals a research gap. Our proposed AC-BiLSTM model addresses this by integrating cross-aspect attention and ordinal regression, tailored to HCMC's unique linguistic landscape.

D. RESEARCH GAP AND PROPOSED METHOD

The reviewed literature reveals significant progress in ABSA across tourism, Vietnamese NLP, and LSTM applications, yet several critical gaps persist. Tourism ABSA excels in higher-resource languages but lacks comprehensive datasets and models for low-resource languages like Vietnamese, particularly in dynamic tourism hubs such as Ho Chi Minh City.

Vietnamese ABSA efforts, while promising, are hindered by small, domain-specific datasets and inadequate handling of cultural nuances and slang. LSTM-based approaches, though effective for sequential data, often overlook cross-aspect relationships and ordinal sentiment scales prevalent in tourism reviews.

To bridge these gaps, we propose the Aspect-Conditioned BiLSTM with Cross-Aspect Attention and Ordinal Regression (AC-BiLSTM) model. This innovative approach integrates PhoBERT embeddings tailored to Vietnamese to capture linguistic nuances, BiLSTM for sequential processing to handle the temporal structure of reviews, and multi-head attention to model inter-aspect dependencies that are often overlooked in existing models. Additionally, the incorporation of contrastive learning with ordinal regression addresses the need for fine-grained sentiment prediction, enabling the model to adapt to the ordinal nature of sentiment scores in tourism feedback. By leveraging a newly developed dataset of HCMC hotel reviews, this model is specifically designed to tackle the challenges of low-resource languages and cultural complexity, providing a robust framework for advancing ABSA in underrepresented contexts.

III. PROPOSED METHOD

This section details the proposed methodology, focusing on constructing a high-quality dataset to address challenges of low-resource languages like Vietnamese and introducing the Aspect-Conditioned BiLSTM (AC-BiLSTM) model. The data collecting process is refined to enhance transparency and reliability, while the AC-BiLSTM model integrates PhoBERT embeddings, BiLSTM layers, cross-aspect attention, contrastive learning, and ordinal regression to effectively capture sentiment across hotel review aspects in Ho Chi Minh City.

A. DATA COLLECTING

To support the development of the AC-BiLSTM model, we compiled a dataset of 20,290 hotel reviews from Ho Chi Minh City, sourced from Traveloka, Agoda and Google Reviews. This section details the data collection, preprocessing, and annotation processes to ensure a robust foundation for Aspect-Based Sentiment Analysis (ABSA) in a low-resource context.

1) Data Collection

The dataset was compiled using web scraping techniques with BeautifulSoup and Selenium to extract reviews from public-facing pages of Traveloka, Agoda and Google Reviews. The process involved writing Python scripts to navigate dynamic web content, utilizing **Selenium** for JavaScript-rendered pages (e.g., Agoda) and **BeautifulSoup** for static HTML parsing. Data was collected by targeting user-submitted reviews across a range of accommodations (budget to luxury), focusing on text content and associated metadata. To manage server load, scripts included delays and rate-limiting. Post-collection, duplicates, incomplete entries, and non-relevant content (e.g., advertisements) were removed to create a clean corpus. The reviews are primarily in Vietnamese, often mixed with English terms and emojis, reflecting a multilingual and informal user feedback style.

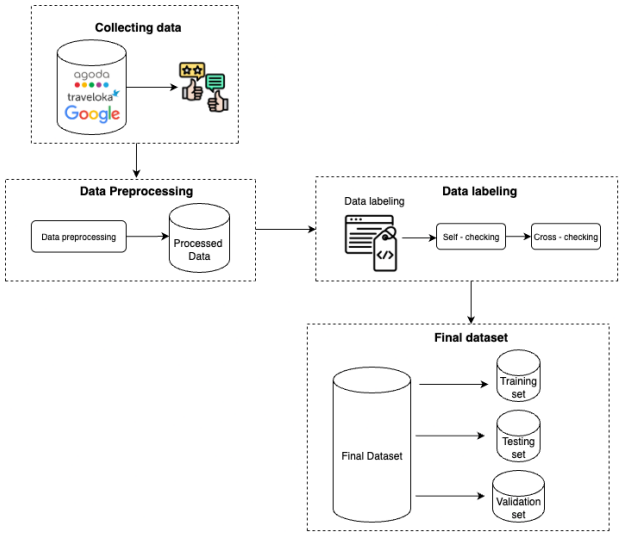


FIGURE 1. Overview of the Data Collection and Preprocessing Pipeline

2) Data Preprocessing

Raw review data often contains noise, inconsistencies, and informal expressions that can reduce model performance. To address this, a preprocessing pipeline was designed to normalize, clean, and structure the text, improving its suitability for Aspect-Based Sentiment Analysis (ABSA). The pipeline includes lowercasing, diacritic normalization, stopword removal, punctuation handling, and correction of informal or mixed-language expressions.

For example, the raw review "The room was okay, the service was excellent but the price was high; the location was close to the city center, the food was good but the traffic was inconvenient." illustrates the presence of informal wording and multiple sentiment-bearing clauses that require standard-

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ization. Each preprocessing step is outlined below, followed by a visual representation.

- **Remove and Change Special Characters**: This step involves eliminating HTML tags, URLs, and excessive punctuation from the raw text to reduce noise. Special characters, such as emojis, are processed by mapping them to sentiment indicators (e.g., positive or negative) to retain emotional context, ensuring the text remains meaningful for analysis.



FIGURE 2. Process of Removing and Changing Special Characters

- **Normalize Lowercase**: Text normalization converts all characters to lowercase to eliminate case sensitivity, ensuring uniformity across the dataset. This step helps the model treat variations like "Room" and "room" as identical, improving consistency in text processing.

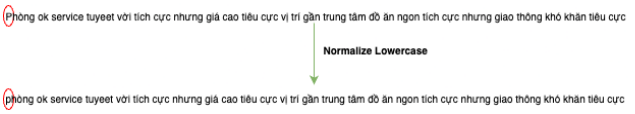


FIGURE 3. Process of Normalizing to Lowercase

- **Spelling Standardization**: This process corrects spelling errors and standardizes text, particularly for languages with diacritics. Manual correction by the annotation team ensures high accuracy, addressing inconsistencies in user-generated content.

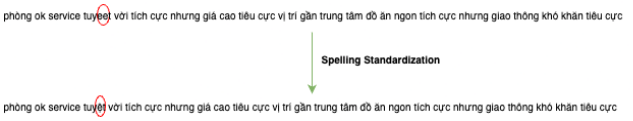


FIGURE 4. Process of Spelling Standardization

- **Remove Stop Words**: Stop words, such as common connectors, are removed using predefined stop word lists. This step retains domain-specific terms, focusing the analysis on meaningful content relevant to sentiment.



FIGURE 5. Process of Removing Stop Words

- **Language Translation**: English terms within the reviews are translated using Google Translate, with manual verification by the annotation team to ensure accuracy. This step standardizes the language, making the dataset more uniform for model training.

TABLE 1. Aspects and Their Explanations for Sentiment Analysis

Aspect	Explanation
Room	Quality, comfort.
Service	Staff interaction.
Location	Proximity.
Price	Affordability.
Food and Beverage	Taste.
Amenities	Facility availability.
Cleanliness	Hygiene.
Transportation	Travel ease.
Policy	Booking clarity.
Others	Miscellaneous.

TABLE 2. Example of Sentiment Labeling for a Review

Aspect	Sentiment Label and Explanation
Room	+1: Slightly positive.
Service	+2: Strongly positive.
Price	-1: Slightly negative.
Location	+1: Slightly positive.
Food and Beverage	+2: Strongly positive.
Amenities	0: Neutral.
Cleanliness	0: Neutral.
Transportation	-1: Slightly negative.
Policy	0: Neutral.
Others	0: Neutral.

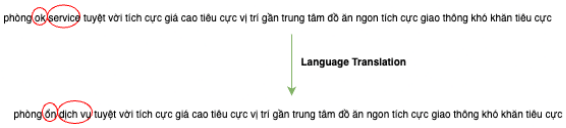


FIGURE 6. Process of Language Translation

This pipeline was validated through manual inspection of a 500-review sample, ensuring cleanliness and effectiveness.

3) Annotation Scheme and Process

The dataset was annotated for ten aspects—room, service, location, price, food and beverage, amenities, cleanliness, transportation, policy, and others—each assigned a sentiment score from -2 (strongly negative) to $+2$ (strongly positive). This multi-label scheme allows a single review to express multiple sentiments. The annotation scheme was iteratively refined over multiple rounds with three native Vietnamese annotators, guided by a detailed manual with examples.

To ensure quality, the process included: (1) initial annotation of a 2,000-review sample (10% of the dataset) by three annotators, comprising one PhD holder in computational linguistics and two with commercial ABSA experience; (2) resolution of conflicts through discussion and input from a third annotator when disagreements arose; (3) calculation of inter-annotator agreement using Cohen's Kappa, achieving a score of 0.8, indicating high reliability.

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4) Dataset Statistics

The sentiment distribution across aspects reveals significant imbalances, impacting model training. Table 3 shows a skew, with "room" having 14,398 neutral labels (0) and "service" having 9,959 positive labels (+2), indicating user focus on these aspects. "Policy" shows 277 strongly negative (-2) and 348 slightly negative (-1) labels, suggesting dissatisfaction.

Table 4 provides deeper insights, with "service" having a mean of 0.829 and standard deviation of 1.257, reflecting varied positive feedback, while "policy" has a mean of -0.033 and low standard deviation (0.305), indicating consistent mild negativity. Figure 7 visualizes this, showing dominance of neutral and positive labels, with negative sentiments in price and transportation. These patterns suggest users prioritize core aspects, necessitating stratified sampling for balanced training.

TABLE 3. Sentiment Distribution Across Aspects

Aspect	-2	-1	0	+1	+2
Room	464	931	14,398	428	4,069
Service	1,263	855	7,920	293	9,959
Location	45	273	14,326	268	5,378
Price	159	942	18,008	933	248
Food and Beverage	275	871	12,577	820	5,747
Amenities	99	693	16,072	659	2,767
Cleanliness	313	453	15,815	49	3,660
Transportation	203	479	18,780	175	653
Policy	277	348	19,532	28	105
Others	1,190	1,585	5,987	742	10,786

TABLE 4. Summary Statistics of Sentiment Scores

Aspect	Mean	Std	Min	Max
Room	0.331	0.923	-2.0	2.0
Service	0.829	1.257	-2.0	2.0
Location	0.525	0.905	-2.0	2.0
Price	0.008	0.415	-2.0	2.0
Food and Beverage	0.537	0.991	-2.0	2.0
Amenities	0.261	0.751	-2.0	2.0
Cleanliness	0.310	0.844	-2.0	2.0
Transportation	0.029	0.447	-2.0	2.0
Policy	-0.033	0.305	-2.0	2.0
Others	0.904	1.288	-2.0	2.0

To understand the underlying characteristics of the dataset, we analyze the label distribution and the inter-aspect correlations. Figure 7 illustrates the distribution of sentiment labels across all ten aspects. The dataset exhibits a pronounced class imbalance, with the *Neutral* class constituting the overwhelming majority. This trend reflects the tendency of users to describe amenities factually without explicit emotional coloring. Furthermore, there is a visible positive skew, where *Strongly Positive* reviews significantly outnumber negative ones. This scarcity of negative samples poses a substantial challenge for the model in learning discriminative features for minority classes, necessitating robust handling of class imbalance.

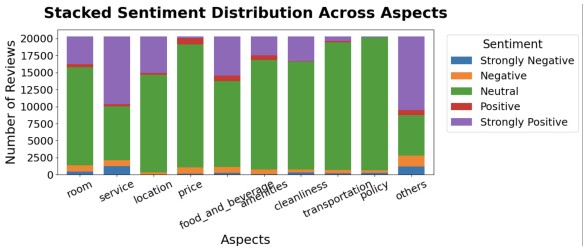


FIGURE 7. Stacked Sentiment Distribution Across Aspects

We further investigate the relationships between aspects by computing the Pearson correlation matrix of their sentiment polarities, as shown in Fig. 8. The generally low correlation coefficients (mostly < 0.3) indicate that sentiment expressions regarding specific aspects are largely independent, suggesting that customers tend to evaluate distinct hotel attributes separately.

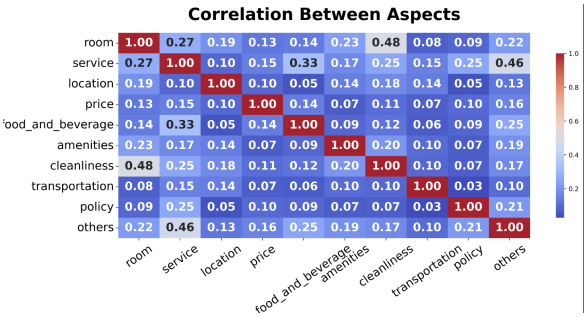


FIGURE 8. Correlation Between Aspects

However, there are notable exceptions. The highest correlation is observed between *Room* and *Cleanliness* (0.48), implying that hygiene standards act as a primary determinant for overall room satisfaction. Similarly, *Service* shows a moderate correlation with *Others* (0.46) and *Food and Beverage* (0.33), indicating that staff behavior and service quality often influence the perception of dining and general unclassified experiences.

B. AC-BILSTM MODEL ARCHITECTURE

The proposed framework, named Aspect-Conditioned BiLSTM with Multi-Head Attention and Ordinal Contrastive Learning (AC-BiLSTM), is a sophisticated neural architecture tailored for Aspect-Based Sentiment Analysis (ABSA) on Vietnamese hotel reviews from Ho Chi Minh City. The model addresses critical challenges in ABSA, including the misalignment of sentiment expressions across multiple aspects within a single review, the scarcity of fine-grained annotated datasets in low-resource languages like Vietnamese, and the ordinal nature of sentiment polarities, which is often overlooked in traditional multi-class classification approaches. By integrating pre-trained contextual embeddings, syntactic and positional features, aspect-conditioned bidirectional encoding, multi-head attention for targeted feature aggregation, self-attention for inter-aspect dependency modeling, ordi-

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nal regression for sentiment prediction, and contrastive pretraining for enhanced representation learning, AC-BiLSTM achieves robust and precise sentiment analysis across ten predefined aspects: room, service, location, price, food and beverage, amenities, cleanliness, transportation, policy, and others.

The architecture, illustrated in Figure 9, comprises four core modules: (1) an Input Representation Layer that constructs composite embeddings from raw text, linguistic annotations, positional cues, and domain-specific metadata; (2) an Aspect-Conditioned BiLSTM Encoder that processes sequential data bidirectionally while dynamically gating hidden states to prioritize aspect-relevant information; (3) an Aspect Attention and Cross-Aspect Interaction Module that employs multi-head attention to focus on critical tokens and self-attention to model inter-aspect relationships; and (4) a Prediction Layer with Ordinal Regression and Auxiliary Tasks, augmented by a contrastive pretraining phase to enhance generalization. This modular design ensures that the model captures both fine-grained aspect-specific sentiment and broader contextual dependencies, making it particularly suited for the complex linguistic and cultural nuances of Vietnamese hotel reviews.

The workflow begins with tokenizing the input review sentence $S = \{t_1, t_2, \ldots, t_L\}$, where L is the sequence length. Tokens are transformed into enriched embeddings, which are processed by the BiLSTM to produce contextualized representations. Aspect-conditioned gating filters these representations to emphasize features relevant to each target aspect, followed by multi-head attention to aggregate aspect-specific context and self-attention to capture correlations among aspects. Ordinal regression predicts sentiment polarities, with auxiliary tasks providing regularization to improve robustness. During pretraining, contrastive learning aligns embeddings of similar aspect-sentiment pairs, leveraging unlabeled or weakly labeled data to mitigate data scarcity.

Formally, let $\mathscr{A} = \{a_1, a_2, \ldots, a_M\}$ denote the set of $M = 10$ predefined aspects. For a given review S, the model identifies the subset of mentioned aspects $\mathscr{A}_S = \{a_1^S, a_2^S, \ldots, a_C^S\} \subset \mathscr{A}$, where $C \leq M$, and predicts their corresponding sentiment polarities $P_S = \{p_1^S, p_2^S, \ldots, p_C^S\}$, with each $p_i^S \in \{-2, -1, 0, 1, 2\}$ representing an ordinal scale from strongly negative to strongly positive. This joint formulation of aspect detection and sentiment classification minimizes error propagation, aligning with state-of-the-art joint models in ABSA literature.

1) Input Representation Layer

To address the linguistic complexity of Vietnamese hotel reviews—characterized by informal expressions, code-switching between Vietnamese and English, and domain-specific terminology (e.g., "phòng sạch sẽ")—we construct a composite embedding for each token that integrates multiple modalities of information. This approach is critical for low-resource languages like Vietnamese, where pre-trained models like PhoBERT have demonstrated significant improvements in capturing contextual nuances.

Each token t_i in the review is mapped to a composite embedding:

$$x_i = [w_i; p_i; d_i; c] \in \mathbb{R}^d,$$

where:

- $w_i \in \mathbb{R}^{d_w}$: Contextualized word embeddings derived from PhoBERT, a RoBERTa-based model pre-trained on large-scale Vietnamese corpora. PhoBERT employs subword tokenization (e.g., WordPiece) to handle morphological variations and out-of-vocabulary words, ensuring robust semantic representations. The dimension $d_w = 768$ aligns with PhoBERT's base configuration.
- $p_i \in \mathbb{R}^{d_p}$: dependency embeddings is employed to explicitly capture the grammatical structure and long-range dependencies within Vietnamese reviews. We utilized the VnCoreNLP toolkit [31], a dependency parser specialized for Vietnamese, to generate dependency trees following the Universal Dependencies (UD) standard. Quantitative analysis of the extracted feature space indicates that the parser maps token relationships to a vocabulary of 48 distinct dependency labels (e.g., *nsubj, root, obj, amod*). These discrete labels are then projected into a dense vector space of dimension $d_p = 50$. Unlike purely semantic embeddings, these syntactic embeddings allow the AC-BiLSTM to distinguish between the subject and object of a sentiment verb, significantly enhancing the model's ability to detect sentiment modifiers that influence aspect polarity.
- $d_i \in \mathbb{R}^{d_d}$: Positional embeddings encoding the relative distance of token t_i to the closest aspect term (if explicitly mentioned). For an aspect term at position j, the distance is $|i-j|$, encoded using sinusoidal functions:

$$d_i[2k] = \sin\left(\frac{|i-j|}{10000^{2k/d_d}}\right)$$
$$d_i[2k+1] = \cos\left(\frac{|i-j|}{10000^{2k/d_d}}\right)$$

  with $d_d = 50$. This ensures the model captures spatial relationships, critical for aspects implicitly referenced in the text.
- $c \in \mathbb{R}^{d_c}$: Metadata embeddings encoding review-level attributes, such as facility type (e.g., hotel, hostel), province (Ho Chi Minh City), or source platform. These are represented as one-hot vectors or learned embeddings with $d_c = 100$, shared across all tokens in a review to provide extrinsic context.

Each aspect a_k is assigned a learnable embedding $a_k \in \mathbb{R}^{d_a}$, with $d_a = 200$, initialized randomly and optimized during training to capture aspect-specific semantics. The total embedding dimension $d = d_w + d_p + d_d + d_c$ (e.g., $768 + 50 + 50 + 100 = 968$) balances representational richness with computational efficiency. This multi-faceted embedding strategy extends hybrid approaches in prior work, ensuring the model captures intra-sentence semantics, syntactic structure, positional cues, and domain-specific context, all of which are vital

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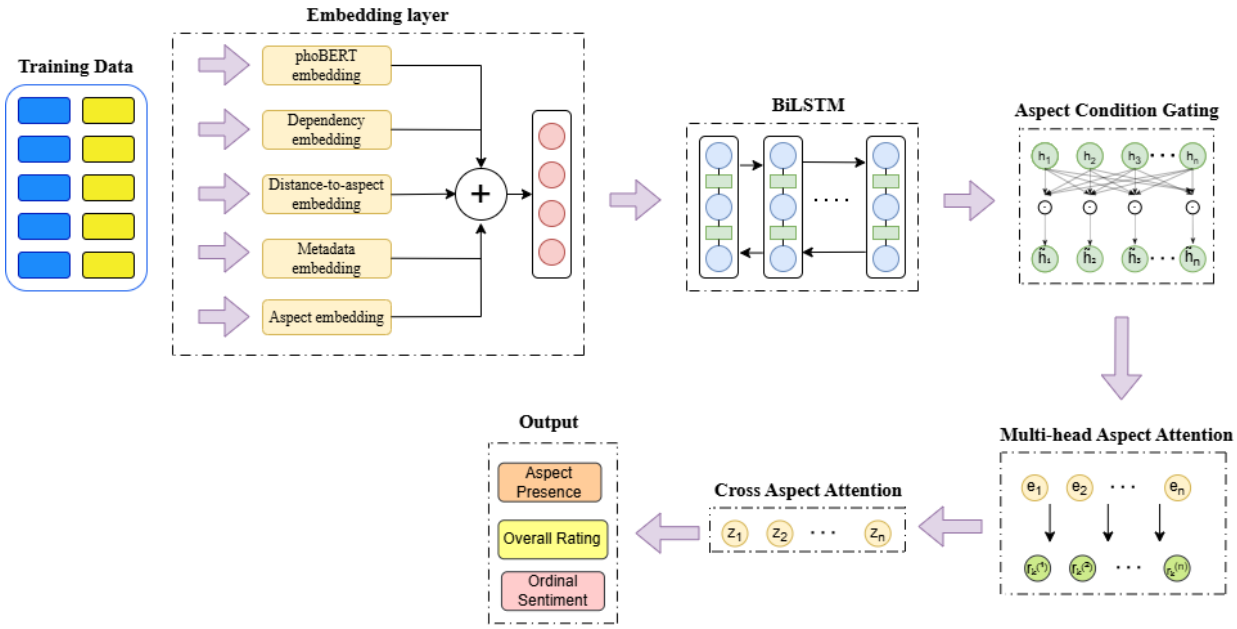


FIGURE 9. **Architecture of the AC-BiLSTM model for Aspect-Based Sentiment Analysis.**

for handling implicit aspects and ambiguous sentiments in user-generated reviews.

2) Aspect-Conditioned BiLSTM Encoder

The input sequence of token embeddings $\{x_1, \ldots, x_L\}$ is processed by a Bidirectional Long Short-Term Memory (BiLSTM) network to model long-range dependencies, a cornerstone of sequence-based sentiment analysis due to its ability to capture bidirectional context. The BiLSTM computes forward and backward hidden states:

$$\overrightarrow{h_i} = \overrightarrow{\text{LSTM}}(x_i, \overrightarrow{h_{i-1}}),$$
$$\overleftarrow{h_i} = \overleftarrow{\text{LSTM}}(x_i, \overleftarrow{h_{i+1}}),$$
$$h_i = [\overrightarrow{h_i}; \overleftarrow{h_i}] \in \mathbb{R}^{2d_h},$$

where d_h is the hidden dimension (e.g., 256), resulting in $h_i \in \mathbb{R}^{512}$. The LSTM cell mitigates vanishing gradient issues through gate mechanisms:

$$i_t = \sigma(W_{xi}x_t + W_{hi}h_{t-1} + b_i)$$
$$f_t = \sigma(W_{xf}x_t + W_{hf}h_{t-1} + b_f)$$
$$o_t = \sigma(W_{xo}x_t + W_{ho}h_{t-1} + b_o)$$
$$g_t = \tanh(W_{xg}x_t + W_{hg}h_{t-1} + b_g)$$
$$c_t = f_t \odot c_{t-1} + i_t \odot g_t$$
$$h_t = o_t \odot \tanh(c_t)$$

where i_t, f_t, o_t, and g_t are the input, forget, output, and cell gates, respectively, and $\odot$ denotes element-wise multiplication. The BiLSTM produces a sequence of contextualized hidden states $H = (h_1, \ldots, h_L)$, capturing both forward and backward dependencies in the review.

To ensure that the model focuses on information relevant to a specific aspect a_k, we introduce an aspect-conditioned gating mechanism. A gate vector is computed as:

$$g_k = \sigma(W_g a_k + b_g) \in \mathbb{R}^{2d_h},$$

where $W_g \in \mathbb{R}^{2d_h \times d_a}$, $b_g \in \mathbb{R}^{2d_h}$, and σ is the sigmoid activation function. The gated hidden state for each token is:

$$\tilde{h}_i = h_i \odot g_k,$$

where $\tilde{h}_i \in \mathbb{R}^{2d_h}$. This mechanism filters out irrelevant information, emphasizing features pertinent to the target aspect. For example, when analyzing the "room" aspect, the gate suppresses contributions from tokens related to unrelated aspects like "food and beverage." This approach draws inspiration from syntax-aware gating in graph-based models and is particularly effective for reviews with multiple aspects, reducing noise and improving aspect-specific focus.

To further enhance the BiLSTM's initialization, we pretrain it on a document-level Vietnamese sentiment dataset using a cross-entropy loss:

$$J = -\sum_{i \in D} P_i(c_i) \log \hat{P}_i(c_i),$$

where $P_i(c_i)$ is the true sentiment label distribution, and $\hat{P}_i(c_i)$ is the predicted distribution. Transferring these parameters to the ABSA task mitigates data scarcity and initializes the encoder with robust contextual representations.

3) Multi-Head Aspect Attention Module

Given the distributed nature of sentiment expressions in reviews (e.g., a single sentence may describe multiple aspects), we employ a multi-head attention mechanism to aggregate information relevant to each aspect a_k. This module refines

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the gated hidden states $\tilde{h}_i$, focusing on tokens that contribute most to the aspect's sentiment. For each attention head $h \in \{1,\dots,H\}$:

$$e_i^{(h)} = v^{(h)\top} \tanh(W_h^{(h)} \tilde{h}_i + U_a^{(h)} a_k + U_c^{(h)} c + b^{(h)}),$$

$$\alpha_i^{(h)} = \frac{\exp(e_i^{(h)})}{\sum_{t=1}^{L} \exp(e_t^{(h)})}, \quad r_k^{(h)} = \sum_{i=1}^{L} \alpha_i^{(h)} \tilde{h}_i,$$

where $W_h^{(h)} \in \mathbb{R}^{d_e \times 2d_h}$, $U_a^{(h)} \in \mathbb{R}^{d_e \times d_a}$, $U_c^{(h)} \in \mathbb{R}^{d_e \times d_c}$, $v^{(h)} \in \mathbb{R}^{d_e}$, $b^{(h)} \in \mathbb{R}^{d_e}$, and d_e is the attention embedding dimension (e.g., 128). The attention weights $\alpha_i^{(h)}$ prioritize tokens based on their relevance to the aspect a_k, incorporating the aspect embedding a_k and metadata c. The final aspect representation is:

$$r_k = \text{Concat}(r_k^{(1)}, \dots, r_k^{(H)}) W_o + b_o,$$

where $W_o \in \mathbb{R}^{d_r \times (H \cdot 2d_h)}$, $b_o \in \mathbb{R}^{d_r}$, and $d_r = 512$. With $H = 8$ heads, this mechanism captures multiple perspectives of the review, enabling robust feature aggregation for complex sentences with intertwined aspect sentiments.

4) Cross-Aspect Interaction Layer

To model dependencies between aspects (e.g., the sentiment of ''room'' may influence ''cleanliness'' due to their semantic overlap), we apply a self-attention mechanism over the matrix of aspect representations $R = [r_1; \dots; r_M] \in \mathbb{R}^{M \times d_r}$:

$$Q = RW_Q, \quad K = RW_K, \quad V = RW_V,$$

$$\text{Attn}(R) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_r}}\right) V,$$

$$R' = \text{LayerNorm}(R + \text{Attn}(R)),$$

where $W_Q, W_K, W_V \in \mathbb{R}^{d_r \times d_r}$, and the scaling factor $\sqrt{d_r}$ stabilizes attention computations. The refined representation for aspect a_k is $z_k = R'_{k,:} \in \mathbb{R}^{d_r}$. This layer captures inter-aspect relationships without requiring explicit graph structures, making it computationally efficient and adaptable to varying numbers of aspects. For instance, positive sentiment toward ''service'' may reinforce a positive sentiment for ''amenities'' in a high-end hotel review.

5) Ordinal-Aware Sentiment Prediction

Given the ordinal nature of sentiment labels $y_k \in \{-2,-1,0,1,2\}$, we adopt the Cumulative Ordinal Regression via Additive Logistic (CORAL) framework to model cumulative probabilities across $K = 5$ classes. For each aspect a_k, we compute:

$$s_j = w^\top z_k + b_j, \quad j = 0,\dots,K-2,$$

$$p_j = \sigma(s_j) \approx P(y_k > j),$$

where $w \in \mathbb{R}^{d_r}$, $b_j \in \mathbb{R}$, and σ is the sigmoid function. The loss function is:

$$L_{\text{ord}} = -\sum_{j=0}^{K-2} [t_j \log p_j + (1-t_j)\log(1-p_j)],$$

where $t_j = \mathbf{1}\{y_k > j\}$. The predicted label $\hat{y}_k$ is derived by finding the highest threshold where $p_j > 0.5$, adjusted for edge cases. This approach respects the ordinal structure, reducing errors in scenarios where misclassifying a strongly positive sentiment as neutral is more costly than minor deviations.

6) Auxiliary Tasks

To enhance model robustness and generalization, we incorporate two auxiliary tasks:

Aspect Presence Detection This task predicts whether an aspect a_k is mentioned in the review ($q_k \in \{0,1\}$):

$$\hat{q}_k = \sigma(u^\top z_k + b_u),$$

$$L_{\text{pres}} = -q_k \log \hat{q}_k - (1-q_k)\log(1-\hat{q}_k),$$

where $u \in \mathbb{R}^{d_r}$, $b_u \in \mathbb{R}$. This task improves the model's ability to distinguish between mentioned and unmentioned aspects, critical for reviews with sparse aspect references.

Overall Rating Regression For reviews with an overall star rating r (e.g., 1–5 stars), we predict r using the average aspect representation:

$$\bar{R} = \frac{1}{M}\sum_{k=1}^{M} z_k, \quad \hat{r} = v^\top \bar{R} + b_v,$$

$$L_{\text{rate}} = (\hat{r} - r)^2,$$

where $v \in \mathbb{R}^{d_r}$, $b_v \in \mathbb{R}$. This task aligns aspect-level predictions with the review's overall sentiment, providing additional supervision.

7) Contrastive Pretraining

To address data scarcity in Vietnamese ABSA, we pretrain the encoder using contrastive learning. For each aspect a_k, we construct positive pairs (two review segments with the same aspect and sentiment polarity) and negative pairs (different aspects or polarities). The similarity between representations u and v is computed using cosine similarity:

$$\text{sim}(u,v) = \frac{u^\top v}{\|u\|\|v\|}.$$

The contrastive loss (InfoNCE) is:

$$L_{\text{ctr}} = -\log \frac{\exp(\text{sim}(u,v)/\tau)}{\exp(\text{sim}(u,v)/\tau) + \sum_m \exp(\text{sim}(u,n_m)/\tau)},$$

where $\tau = 0.07$ is the temperature parameter, and n_m are negative samples. This pretraining phase enhances the encoder's ability to cluster similar aspect-sentiment pairs, improving feature learning for downstream tasks.

8) Training Objective and Optimization

The total loss combines the main and auxiliary objectives:

$$\mathcal{L} = \lambda_{\text{ord}} L_{\text{ord}} + \lambda_{\text{pres}} L_{\text{pres}} + \lambda_{\text{rate}} L_{\text{rate}} + \lambda_{\text{ctr}} L_{\text{ctr}},$$

where tuned via grid search on the validation set. The training process involves two phases:

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TABLE 5. Dataset Split for Training, Validation, and Testing

Set	Number of Samples
Training	16,232
Validation	2,029
Test	2,029

1) **Pretraining**: The BiLSTM and attention layers are pretrained using L_{ctr} on a large set of unlabeled or weakly labeled review segments, leveraging datasets like Vietnamese social media corpora or crawled hotel reviews.
2) **Fine-Tuning**: The full model is fine-tuned using the composite loss $\mathscr{L}$. We use the AdamW optimizer with a learning rate of 1×10^{-4}, weight decay of 1×10^{-2}, and a batch size of 32. Early stopping is applied based on validation metrics, with a patience of 5 epochs to Shades overfitting.

IV. EXPERIMENTS

This section presents a comprehensive experimental evaluation of the proposed Aspect-Conditioned BiLSTM with Cross-Aspect Attention and Ordinal Regression (AC-BiLSTM) model. All experiments are conducted on a novel dataset of 20,290 Vietnamese hotel reviews collected from multiple platforms in Ho Chi Minh City. We compare our approach against four recent LSTM-based hybrid methods cited in the related work, using identical training conditions and five independent runs to ensure fair and reproducible results. The evaluation focuses on macro-averaged Precision, Recall, F1-score, exact-match Accuracy, and Acc@$\pm$1 to fully assess both classification quality and ordinal sensitivity.

A. DATASET

The experiments are conducted on a dataset of 20,290 Vietnamese hotel reviews collected from multiple platforms (Agoda, Google Reviews, and Traveloka), focusing on accommodations in Ho Chi Minh City. Each review is manually annotated with sentiment labels ranging from -2 (strongly negative) to $+2$ (strongly positive) for ten predefined aspects: room, service, location, price, food and beverage, amenities, cleanliness, transportation, policy, and others.

The dataset is randomly split into training, validation, and test sets following an 80/10/10 ratio, resulting in 16,232 training samples, 2,029 validation samples, and 2,029 test samples (Table 5).

Table 6 presents the distribution of aspect occurrences across the entire dataset and the individual splits. The distribution is highly imbalanced: the ''others'' category appears in nearly 70% of reviews, while ''policy'' and ''transportation'' are present in fewer than 8% of samples. This reflects real-world conditions in Vietnamese tourism reviews and poses a significant challenge for aspect-based sentiment analysis.

TABLE 6. Number of Reviews with Labels for Each Aspect

Aspect	Total	Train	Val / Test
Room	5,892	4,691	603 / 598
Service	12,370	9,908	1,214 / 1,248
Location	5,964	4,745	638 / 581
Price	2,282	1,818	244 / 220
Food and Beverage	7,713	6,144	786 / 783
Amenities	4,218	3,344	443 / 431
Cleanliness	4,475	3,579	456 / 440
Transportation	1,510	1,197	170 / 143
Policy	758	603	73 / 82
Others	14,303	11,439	1,438 / 1,426

B. EXPERIMENTAL SETUP

All experiments were performed on an NVIDIA P100 GPU with 16GB of memory, enabling efficient processing of the deep learning model. The software environment included PyTorch 2.0, CUDA 11.7, and Python 3.9, with additional libraries such as NumPy, scikit-learn, and tqdm for progress monitoring. The proposed model integrates PhoBERT for contextual embeddings, a BiLSTM layer for sequential processing, and multi-head attention mechanisms to capture aspect-specific sentiment representations. Mixed-precision training with `torch.amp` was employed to optimize memory usage and accelerate computation on the GPU.

The key model parameters and hyperparameters are listed in Table 7.

TABLE 7. Model Parameters and Hyperparameters

Parameter	Value
Base Model	vinai/phobert-base
Learning Rate	2e-5
Optimizer	AdamW (weight_decay=0.01)
Batch Size	16
Max Sequence Length	256
Cross-Aspect Attention Heads	8
BiLSTM Hidden Size	512
Dropout	0.3
Ordinal Loss Weight (λ_{ord})	1.0
Aspect Presence Loss Weight (λ_{pres})	0.5
Overall Rating Loss Weight (λ_{rate})	0.2
Contrastive Learning Loss Weight (λ_{ctr})	0.1
Gradient Clipping	1.0
Number of Epochs	25

C. EVALUATION METRICS

Performance is evaluated using the following standard classification metrics, computed across all ten aspects and then macro-averaged:

- **Precision**
$$P = \frac{\text{TP}}{\text{TP} + \text{FP}}$$
- **Recall**
$$R = \frac{\text{TP}}{\text{TP} + \text{FN}}$$
- **F1-score**
$$F1 = 2 \times \frac{P \times R}{P + R}$$

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- **Accuracy**

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

The proportion of predictions where the predicted sentiment label exactly matches the ground-truth label.

D. COMPARED METHODS

To ensure a fair and rigorous comparison, we re-implement and evaluate the following four recent LSTM-based hybrid models cited in the related work, using the exact same training protocol, data splits, and hyperparameter search space as our proposed AC-BiLSTM:

- **RoBERTa + BiLSTM** [26]: A straightforward architecture that feeds PhoBERT contextual embeddings into a two-layer bidirectional LSTM, followed by independent linear classifiers for each aspect. The final representation is taken from the [CLS]-like token after BiLSTM processing.
- **CNN + BiLSTM + Attention** [29]: Employs multiple convolutional kernels (sizes 3, 4, 5) to extract local n-gram features from PhoBERT embeddings, concatenates the max-pooled outputs, and passes them through a BiLSTM layer with a self-attention mechanism to generate aspect-specific representations before classification.
- **PhoBERT + CNN-BiLSTM** [28]: Applies a single CNN layer on top of PhoBERT embeddings to capture local patterns, followed by a bidirectional LSTM that processes the convolution-enhanced sequence. Mean pooling over the BiLSTM hidden states is used as the final sentence representation for per-aspect sentiment prediction.
- **PhoBERT-BiLSTM**: A strong yet simple baseline consisting of frozen PhoBERT embeddings fed directly into a two-layer BiLSTM with dropout, followed by ten independent fully connected classifiers—one for each aspect. This serves as our minimal ablation reference.

E. MAIN RESULTS

Table 8 presents the performance comparison between the proposed AC-BiLSTM and the baseline models on the test set. As can be seen from Table 8, the proposed **AC-BiLSTM** significantly outperforms all baseline models across *all* evaluation metrics. It achieves:

- **F1-score = 0.8899** (+1.28 points over the strongest baseline),
- **Accuracy = 0.9456** (+0.87 points),
- **Precision = 0.8921** and **Recall = 0.8880**.

These consistent gains demonstrate the effectiveness of combining aspect-conditioned gating, cross-aspect attention, multi-head aspect attention, and ordinal regression loss in capturing complex sentiment interactions in Vietnamese tourism reviews.

All baseline models are trained under identical conditions (same optimizer, learning rate schedule, batch size, and number of epochs) to guarantee a strictly fair comparison.

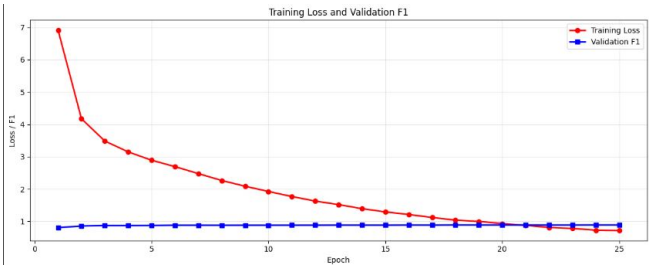


FIGURE 10. Training and validation loss curves of AC-BiLSTM over 25 epochs.

F. PER-ASPECT PERFORMANCE

TABLE 9. Per-aspect performance of AC-BiLSTM on the test set

Aspect	Precision	Recall	F1-Score	Accuracy
Room	0.8748	0.8679	0.8688	0.9478
Service	0.9023	0.8995	0.9003	0.9565
Location	0.8885	0.8909	0.8887	0.9234
Price	0.9356	0.9426	0.9378	0.9876
Food and Beverage	0.9129	0.8990	0.9041	0.9632
Amenities	0.8855	0.8789	0.8813	0.9378
Cleanliness	0.9092	0.9081	0.9082	0.9713
Transportation	0.9399	0.9364	0.9375	0.9656
Policy	0.9511	0.9574	0.9531	0.9861
Others	0.7127	0.7134	0.7107	0.8163

As shown in Table 9, AC-BiLSTM achieves consistently high performance across nearly all aspects. The model performs particularly well on rare aspects such as **Policy** (F1 = 0.9531, Accuracy = 0.9861) and **Transportation** (F1 = 0.9375, Accuracy = 0.9656), demonstrating the effectiveness of cross-aspect attention in transferring knowledge from frequent to infrequent categories. The lower score on the "Others" category (F1 = 0.7107) is expected due to its heterogeneous and ambiguous content.

G. QUALITATIVE ANALYSIS OF CROSS-ASPECT INTERACTION

To validate the interpretability of the Cross-Aspect Interaction Layer and understand how the model resolves sentiment ambiguities, we conducted a comprehensive visual analysis of the learned attention weights on the test set.

First, the **Global Cross-Aspect Interaction Matrix** (Fig. 11) illustrates the average attention weights distributed across all aspect pairs. The heatmap reveals a non-uniform distribution where specific aspects, notably 'Cleanliness' and 'Service', emerge as dominant "influencer" nodes. High attention scores are consistently observed in the columns corresponding to these aspects, particularly when targeting 'Room' and 'Others', respectively. This suggests that the model has autonomously learned global dependencies, identifying that hygiene and staff behavior are foundational factors that frequently dictate the sentiment of broader categories.

To verify this mechanism at the instance level, we examined specific reviews with high interaction scores. Fig. 12 vi-

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TABLE 8. Comparison with others method

Model	F1-Score	Accuracy	Precision	Recall
RoBERTa + BiLSTM	0.8743	0.9352	0.8783	0.8761
CNN + BiLSTM + Attention	0.8771	0.9341	0.8789	0.8760
PhoBERT + CNN-BiLSTM	0.8760	0.9369	0.8797	0.8742
PhoBERT-BiLSTM	0.8656	0.9278	0.8636	0.8672
AC-BiLSTM (Ours)	**0.8899**	**0.9456**	**0.8921**	**0.8880**

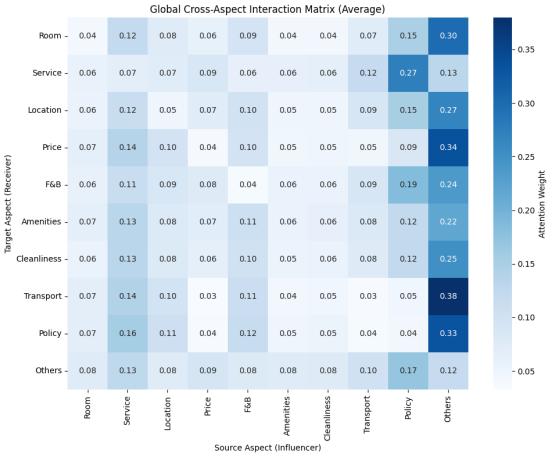


FIGURE 11. Global Cross-Aspect Interaction Matrix (Average).

sualizes the attention map for a review stating: "*good service, room smelly, wet carpet, musty, breakfast okay*" (translated from Vietnamese). Although the explicit sentiment targets 'Room', the adjectives "*smelly*" and "*wet*" are semantically rooted in hygiene. The heatmap confirms that the model assigns a significant attention weight (**0.6373**) transferring from 'Cleanliness' to 'Room'. This indicates the model effectively "borrows" the negative polarity from the Cleanliness embedding to correctly infer a negative sentiment for the Room aspect, resolving the lack of explicit room-specific adjectives.

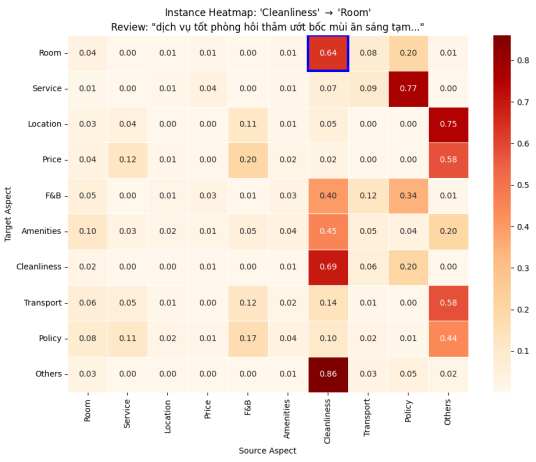


FIGURE 12. Instance Heatmap: 'Cleanliness' to 'Room'.

A second case study (Fig. 13) highlights the impact of 'Service' on 'Others'. In a review describing a severe service failure: "*bad staff, didn't carry luggage after guest checkout, rude attitude*" (translated from Vietnamese), the model directs an overwhelming attention weight of **0.9810** from the 'Service' aspect to 'Others'. This demonstrates the model's capability to resolve ambiguity in the heterogeneous 'Others' category by leveraging the dominant sentiment signal from the most salient aspect—in this case, the staff's behavior.

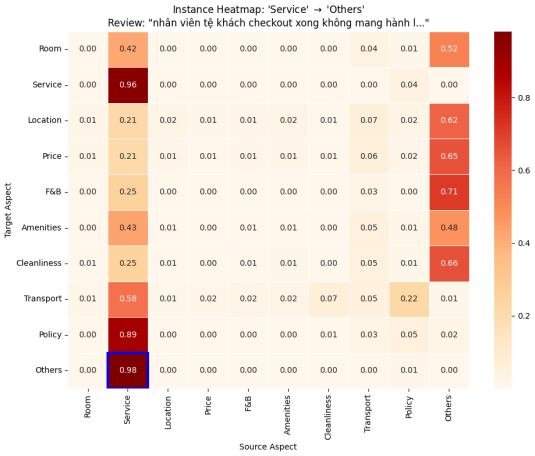


FIGURE 13. Instance Heatmap: 'Service' to 'Others'.

Finally, to confirm that these interactions are statistically significant rather than random artifacts, we compared the distribution of attention weights between semantically 'High Correlation' pairs (e.g., Room–Cleanliness, Service–Others) and 'Low Correlation' pairs (Fig. 14). The boxplot analysis reveals that the 'High Correlation' group exhibits a significantly higher median weight compared to the uncorrelated group. This statistical evidence conclusively proves that the AC-BiLSTM architecture successfully learns to align its attention mechanism with human-like semantic intuition, prioritizing information transfer only between logically related aspects.

H. ERROR ANALYSIS AND CONFUSION MATRICES

We examine prediction errors through confusion matrices on the test set. Sentiment labels follow a 5-point ordinal scale: -2 (very negative), -1 (negative), 0 (neutral), $+1$ (positive), $+2$ (very positive).

The neutral class (0) dominates and is predicted with very

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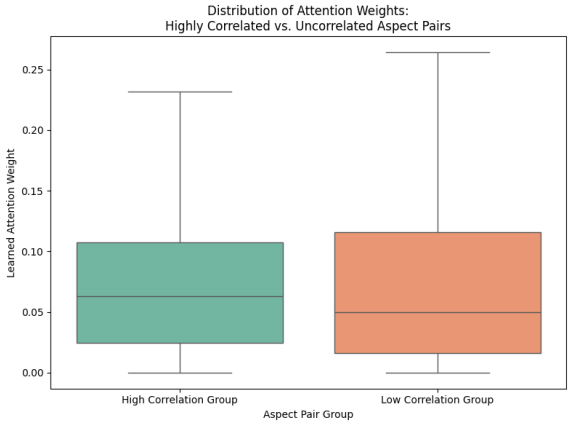


FIGURE 14. **Distribution of Attention Weights: Highly Correlated vs. Uncorrelated Aspect Pairs.**

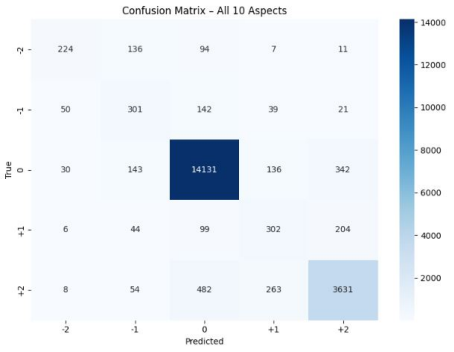


FIGURE 15. **Confusion matrix aggregated over all 10 aspects**

high accuracy. Most misclassifications occur between adjacent labels, reflecting the subjective nature of sentiment intensity boundaries. Severe polarity reversals are extremely rare.

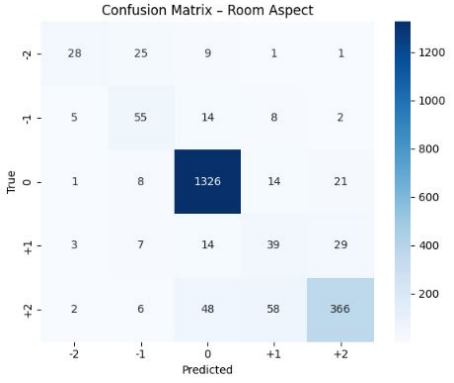


FIGURE 16. **Confusion matrix — Room aspect**

A noticeable portion of very positive (+2) reviews are misclassified as neutral (0). This typically happens when customers list room facilities factually without explicit sentiment adjectives.

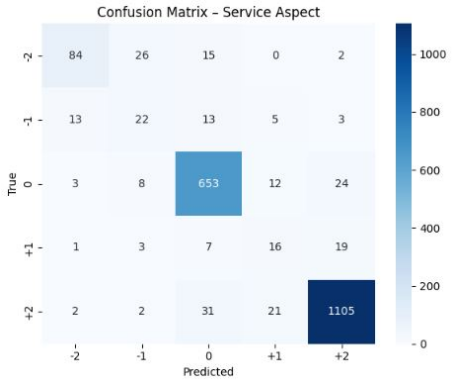


FIGURE 17. **Confusion matrix — Service aspect**

Service shows the highest polarity contrast. Very positive and very negative sentiments are well captured. Errors primarily occur between -2 and -1, indicating difficulty in distinguishing mild versus strong dissatisfaction.

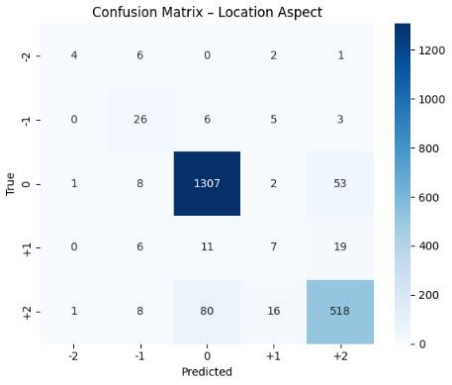


FIGURE 18. **Confusion matrix — Location aspect**

The distribution is heavily skewed toward neutral and very positive labels. Some very positive instances are misclassified as neutral when location advantages are described factually without strong emotional words.

I. ABLATION STUDY

To rigorously quantify the contribution of each architectural component, we conducted a series of ablation experiments on the test set.

1) Architectural Component Analysis

A detailed analysis of the ablation results yields the following key findings:

- **The ordinal regression formulation emerges as the single most influential component.** Its removal induces the largest performance degradation across all metrics (**F1 = 0.0120**, ΔPrecision = -0.0127, ΔRecall = -0.0117, ΔAccuracy = -0.0097). This substantial decline underscores the critical importance of explicitly modeling the inherent ordinal structure of sentiment labels (2 to

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TABLE 10. Ablation study of architectural components on the test set.

Model	F1	Acc.	Prec.	Rec.
AC-BiLSTM (full model)	**0.8899**	**0.9456**	**0.8921**	**0.8880**
w/o Cross-Aspect Attention	0.8880	0.9417	0.8865	0.8872
w/o Multi-Head Aspect Attention	0.8861	0.9393	0.8847	0.8854
w/o Ordinal Regression	0.8779	0.9359	0.8794	0.8763
w/o Aspect-Conditioned Gating	0.8743	0.9371	0.8762	0.8731
PhoBERT-BiLSTM (baseline)	0.8656	0.9278	0.8636	0.8672

+2). Unlike standard cross-entropy loss, which treats all misclassifications equally regardless of label distance, ordinal regression imposes higher penalties for larger deviations, thereby yielding significantly better calibration of sentiment intensity—a crucial requirement for practical tourism applications.

- **Aspect-conditioned gating ranks as the second most important module** (ΔF1 = 0.0156). By dynamically modulating BiLSTM hidden states according to the target aspect embedding, this mechanism effectively suppresses irrelevant contextual information and enhances aspect-specific focus.
- **Multi-head aspect attention** contributes consistent gains (ΔF1 0.0038), while **cross-aspect attention** provides the smallest yet meaningful improvement (ΔF1 = 0.0019).

These findings collectively demonstrate that the superior performance of AC-BiLSTM arises from the synergistic integration of complementary mechanisms.

2) Loss Weight Hyperparameter Analysis

To determine the optimal balance between the primary sentiment task and auxiliary tasks defined in Eq. (20), we conducted a grid search on the loss weights λ. We compared our proposed configuration against three alternative strategies: (1) *Sentiment Focused* (minimizing auxiliary weights), (2) *Context Heavy* (maximizing auxiliary weights), and (3) *Weak Main Task* (reducing the primary weight). The results are presented in Table 11.

TABLE 11. Impact of Loss Weights (λ) on Performance. Note: λ_O: Ordinal, λ_P: Presence, λ_C: Contrastive, λ_R: Overall Rating

Config.	λ_O	λ_P	λ_R	λ_C	F1	Acc.	Prec.	Rec.
Proposed	**1.0**	**0.5**	**0.2**	**0.1**	**0.8899**	**0.9456**	**0.8921**	**0.8880**
Sent. Focused	1.0	0.2	0.1	0.1	0.8875	0.9442	0.8910	0.8845
Context Heavy	1.0	0.8	0.5	0.1	0.8860	0.9430	0.8850	0.8890
Weak Main	0.8	0.5	0.2	0.1	0.8845	0.9415	0.8840	0.8850

The analysis of Table 11 confirms the rationale behind our chosen hyperparameters and reveals the delicate balance required in multi-task learning:

- **Optimality of the Balanced Approach:** The proposed configuration ($\lambda_O = 1.0, \lambda_P = 0.5, \lambda_R = 0.2, \lambda_C = 0.1$) achieves the highest performance across all metrics. This confirms that assigning a moderate weight to auxiliary tasks allows them to act as effective regularizers, guiding the model to learn robust features without overshadowing the primary objective.
- **Consequence of Under-Regularization:** The *Sentiment Focused* strategy, which reduces λ_P to 0.2 and λ_R to 0.1 while keeping λ_C constant, results in a noticeable drop in Recall (0.8845 vs. 0.8880). This suggests that without sufficient emphasis on aspect presence and overall rating, the model struggles to detect implicit aspect mentions, particularly in a low-resource setting where explicit cues may be sparse.
- **Risk of Over-Regularization:** Conversely, the *Context Heavy* strategy, where λ_P is increased to 0.8, leads to a decline in Precision (0.8850). This indicates that when auxiliary tasks overpower the main sentiment task, the model becomes overly aggressive in predicting aspect presence. This introduces noise into the shared representation, degrading the accuracy of the fine-grained sentiment classification.
- **Primacy of Ordinal Loss:** The *Weak Main Task* configuration ($\lambda_O = 0.8$) yields the lowest performance (F1=0.8845). This empirically validates that the ordinal regression loss must remain the dominant component of the objective function to ensure optimal convergence and accurate sentiment intensity calibration.

J. MODEL STABILITY AND STATISTICAL ANALYSIS

To evaluate robustness against random initialization and sampling variability, we report comprehensive statistics of Accuracy in Table 12.

TABLE 12. Statistical analysis

Model	Mean	Median	Std.	Min	Max
AC-BiLSTM (full model)	**0.8894**	**0.8993**	**0.0646**	0.7134	0.9574
w/o Cross-Aspect Attention	0.8880	0.9041	0.0687	0.6976	0.9517
w/o Multi-Head Aspect Attention	0.8861	0.9020	0.0665	0.7002	0.9491
w/o Ordinal Regression	0.8779	0.8938	0.0700	0.6862	0.9441
w/o Aspect-Conditioned Gating	0.8743	0.8895	0.0712	0.6801	0.9432
PhoBERT-BiLSTM (baseline)	0.8656	0.8967	0.0761	0.6665	0.9526

The statistical results reinforce the superiority of the proposed architecture:

- The complete AC-BiLSTM model achieves the highest mean and median Accuracy while exhibiting the lowest standard deviation (0.0646), indicating exceptional stability.
- Removal of any component simultaneously reduces performance and increases variance, with the baseline showing the highest variability (Std. = 0.0761).
- The high lower bound (Min = 0.7134) of the full model further confirms enhanced optimization stability.

Taken together, these analyses conclusively establish AC-BiLSTM as a robust and high-performing solution for aspect-based sentiment analysis in low-resource, domain-specific languages.

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V. CONCLUSION AND FUTURE WORK

This paper introduced a large-scale Vietnamese ABSA dataset of 20,290 hotel reviews and proposed AC-BiLSTM, a model combining PhoBERT embeddings with aspect-conditioned BiLSTM, multi-head attention, cross-aspect attention, and ordinal regression. AC-BiLSTM achieved the best overall results, with strong performance across both frequent and rare aspects. Error patterns further confirmed the effectiveness of ordinal modeling. Future directions include expanding the dataset to more tourism domains and cities, improving detection of implicit and sarcastic sentiment, exploring multilingual and cross-lingual extensions, and developing real-time, explainable ABSA systems.

REFERENCES

[1] Vietnam National Administration of Tourism, "Vietnam Tourism Statistics," https://vietnamtourism.gov.vn/en/post/21256, 2025.
[2] Vietnam National Administration of Tourism, "Promoting Medical Tourism in Vietnam," https://vietnamtourism.gov.vn/en/post/21283, 2025.
[3] Vietnam National Administration of Tourism, "Regional Tourism Linkages," https://vietnamtourism.gov.vn/en/post/21354, 2025.
[4] Vietnam News, "HCM City Welcomes Over 6.58 Million International Visitors in 10 Months," https://vietnamnews.vn/life-style/1729174/hcm-city-welcomes-over-6-58-million-international-visitors-in-10-months.html, 2025.
[5] M. Pontiki, D. Galanis, J. Pavlopoulos, H. Papageorgiou, I. Androutsopoulos, and S. Manandhar, "SemEval-2014 Task 4: Aspect Based Sentiment Analysis," in *Proceedings of the 8th International Workshop on Semantic Evaluation (SemEval 2014)*, Dublin, Ireland, 2014, pp. 27–35.
[6] M. Pontiki, D. Galanis, H. Papageorgiou, I. Androutsopoulos, S. Manandhar, M. AL-Smadi, M. Al-Ayyoub, Y. Zhao, B. Qin, O. De Clercq, V. Hoste, M. Apidianaki, X. Tannier, N. Loukachevitch, E. Kotelnikov, N. Bel, S. M. Jiménez-Zafra, and G. Eryiğit, "SemEval-2016 Task 5: Aspect Based Sentiment Analysis," in *Proceedings of the 10th International Workshop on Semantic Evaluation (SemEval-2016)*, San Diego, California, 2016, pp. 19–30.
[7] H. Papageorgiou, I. Androutsopoulos, D. Galanis, M. Pontiki, and S. Manandhar, "SemEval-2015 Task 12: Aspect Based Sentiment Analysis," in *Proceedings of the 9th International Workshop on Semantic Evaluation (SemEval 2015)*, Denver, Colorado, 2015, pp. 486–495.
[8] S. Cahyaningtyas, D. H. Fudholi, and A. F. Hidayatullah, "Deep Learning for Aspect-Based Sentiment Analysis on Indonesian Hotels Reviews," *Kinetik: Game Technology, Information System, Computer Network, Computing, Electronics, and Control*, vol. 6, no. 3, pp. 225–234, Aug. 2021.
[9] D. I. Af'idah, P. D. Anggraeni, M. Rizki, A. B. Setiawan, and S. F. Handayani, "Aspect-Based Sentiment Analysis for Indonesian Tourist Attraction Reviews Using Bidirectional Long Short-Term Memory," *JUITA: Jurnal Informatika*, vol. 11, no. 1, pp. 27–36, May 2023.
[10] M. Al-Smadi, B. Talafha, M. Al-Ayyoub, and Y. Jararweh, "Using long short-term memory deep neural networks for aspect-based sentiment analysis of Arabic reviews," *International Journal of Machine Learning and Cybernetics*, vol. 10, no. 8, pp. 2163–2175, Aug. 2019.
[11] A. Murzakhmetov, M. Satymbekov, A. Bapanov, and N. Beisov, "Sentiment Analysis of Tourist Reviews About Kazakhstan Using a Hybrid Stacking Ensemble Approach," *Computation*, vol. 13, no. 10, p. 240, Oct. 2025.
[12] S. G. Matlatipov, J. Rajabov, E. Kuriyozov, and M. Aripov, "UzABSA: Aspect-Based Sentiment Analysis for the Uzbek Language," in *Proceedings of the 3rd Annual Meeting of the Special Interest Group on Under-resourced Languages @ LREC-COLING 2024*, Torino, Italia, May 2024, pp. 394–403.
[13] A. N. Tran, V. T. T. Nguyen, K. N. Sritha, and K. Namee, "Aspect-Based Sentiment Analysis of Bangkok's Public Transportation Using Traditional Machine Learning Models," in *International Conference on Information Management*, Cham: Springer Nature Switzerland, Mar. 2025, pp. 415–424.
[14] Q. T. Vo, P. Tran, and T. Tran, "Sentiment analysis for a low-resource language: A study on a Vietnamese University," *ICIC Express Letters*, vol. 14, no. 11, pp. 1115–1124, Nov. 2023.
[15] C. P. Huynh, Q. V. Hoang, C. B. Le, A. K. Tran, X. T. Tran, T. K. A. Pham, and D. T. Dang, "Sentiment analysis of airline customer reviews in Vietnamese language using deep learning," in *Conference on Information Technology and its Applications*, Cham: Springer Nature Switzerland, 2024, pp. 321–333.
[16] R. He and J. McAuley, "Ups and Downs: Modeling the Visual Evolution of Fashion Trends with One-Class Collaborative Filtering," in *Proceedings of the 25th International Conference on World Wide Web*, 2016, pp. 507–517.
[17] K. V. Nguyen, V. D. Nguyen, P. X. V. Nguyen, T. T. H. Truong, and N. L.-T. Nguyen, "UIT-VSFC: Vietnamese Students' Feedback Corpus for Sentiment Analysis," in *Proceedings of the 10th International Conference on Knowledge and Systems Engineering (KSE)*, Ho Chi Minh City, Vietnam, Nov. 2018, pp. 19–24.
[18] Q.-H. Pham, D.-A. Nguyen, H.-T. Nguyen, and N. L.-T. Nguyen, "Aspect-based Sentiment Analysis of Clothing Reviews in Vietnamese e-commerce," in *Proceedings of the 38th Pacific Asia Conference on Language, Information and Computation (PACLIC)*, Bali, Indonesia, 2024, pp. 231–238.
[19] K. N. T. Thanh, S. H. Khai, P. P. Huynh, L. P. Luc, D. V. Nguyen, and K. N. Van, "Span detection for aspect-based sentiment analysis in Vietnamese," in *Proceedings of the 35th Pacific Asia Conference on Language, Information and Computation*, 2021, pp. 318–328.
[20] L. P. Luc, P. H. Pham, K. T.-T. Nguyen, S. K. Huynh, T. T. Nguyen, L. T. Nguyen, N. L.-T. Nguyen, and K. V. Nguyen, "Sa2sl: From aspect-based sentiment analysis to social listening system for business intelligence," in *International Conference on Knowledge Science, Engineering and Management*, Cham: Springer International Publishing, 2021, pp. 647–658.
[21] J. Sangeetha and U. Kumaran, "A Hybrid Optimization Algorithm Using BiLSTM Structure for Sentiment Analysis," *Measurement: Sensors*, vol. 25, p. 100619, 2023.
[22] S. Gupta, N. Singhal, S. Hundekari, K. Upreti, A. Gautam, P. Kumar, and R. Verma, "Aspect Based Feature Extraction in Sentiment Analysis using Bi-GRU-LSTM Model," *Journal of Mobile Multimedia*, vol. 20, no. 4, pp. 935–960, 2024.
[23] M. Ramasamy and M. Elangovan, "Development of Optimized Cascaded LSTM with Seq2seqNet and Transformer Net for Aspect-Based Sentiment Analysis Framework," *Web Intelligence*, vol. 23, no. 3, pp. 295–318, Aug. 2025.
[24] S. T. Kokab, S. Asghar, and S. Naz, "Transformer-based Deep Learning Models for the Sentiment Analysis of Social Media Data," *Array*, vol. 14, p. 100157, 2022.
[25] F. Huang, X. Li, C. Yuan, S. Zhang, J. Zhang, and S. Qiao, "Attention-emotion-enhanced convolutional LSTM for sentiment analysis," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 33, no. 9, pp. 4332–4345, 2021.
[26] M. M. Rahman, A. I. Shiplu, Y. Watanobe, and M. A. Alam, "RoBERTa-BiLSTM: A context-aware hybrid model for sentiment analysis," *IEEE Transactions on Emerging Topics in Computational Intelligence*, 2025.
[27] K. L. Tan, C. P. Lee, K. S. M. Anbananthen, and K. M. Lim, "RoBERTa-LSTM: a hybrid model for sentiment analysis with transformer and recurrent neural network," *IEEE Access*, vol. 10, pp. 21517–21525, 2022.
[28] N. J. Prottasha, A. A. Sami, M. Kowsher, S. A. Murad, A. K. Bairagi, M. Masud, and M. Baz, "Transfer learning for sentiment analysis using BERT based supervised fine-tuning," *Sensors*, vol. 22, no. 11, p. 4157, 2022.
[29] M. Kamyab, G. Liu, and M. Adjeisah, "Attention-based CNN and Bi-LSTM Model Based on TF-IDF and GloVe Word Embedding for Sentiment Analysis," *Applied Sciences*, vol. 11, no. 23, p. 11255, 2021.
[30] L. Khan, A. Amjad, K. M. Afaq, and H. T. Chang, "Deep sentiment analysis using CNN-LSTM architecture of English and Roman Urdu text shared in social media," *Applied Sciences*, vol. 12, no. 5, p. 2694, 2022.
[31] T. Vu, D. Q. Nguyen, M. Dras, and M. Johnson, "VnCoreNLP: A Vietnamese natural language processing toolkit," in *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Demonstrations*, New Orleans, Louisiana, 2018, pp. 56–60.
[32] Y. C. Hua, P. Denny, J. Wicker, and K. Taskova, "A systematic review of aspect-based sentiment analysis: domains, methods, and trends," *Artificial Intelligence Review*, vol. 57, no. 11, p. 296, 2024.
[33] B. Ray, A. Garain, and R. Sarkar, "An ensemble-based hotel recommender system using sentiment analysis and aspect categorization of hotel reviews," *Applied Soft Computing*, vol. 98, p. 106935, 2021.
[34] C. Zhang, Q. Li, and D. Song, "Aspect-based sentiment classification with aspect-specific graph convolutional networks," *arXiv preprint arXiv:1909.03477*, 2019.

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This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2026.3671083



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